LYRICS → MODEL → MUSIC / A LOCAL ADAPTATION STUDY

KAZAKH WORDS. NEW SOUND.

We built a local prompt-to-music workflow and trained an adapter on Kazakh music and singing.

Yeldar · Aisha · Dimash

Kazakh lyrics-conditioned music generation

OUR BUILD JOURNEY / TWO DIFFERENT DATA PROBLEMS

FIRST THE DOMBRA. THEN THE VOICE.

01

Listen and classify

We reviewed 116 archive recordings by ear: solo dombra, voice with dombra, or orchestra. Labels alone were not enough.

02

Instrumental expansion

We used 59 dombra-only recordings to create 111 new instrumental crops. This stage focused on the sound of the instrument.

03

A separate vocal workflow

Vocals also needed words aligned to each clip. We held 27 dombra-and-voice recordings back while their lyrics remained unverified.

These were successive experiments and separate preparation tasks, not two models whose outputs we mixed together. The earlier pilot already contained a few vocal crops.

01	02	03
ONLINE VIDEO Wikimedia Commons	ARCHIVE MUSIC RTRK recordings	ALIGNED VOCALS Hugging Face

Our recording-by-recording review ↗ Video and dataset source trail ↗ Audio and dataset checks ↗

THE VOCAL STAGE / DATA FROM HUGGING FACE

HUGGING FACE.
ALIGNED KAZAKH VOCALS.

3,013

source segments

195 songs · 36 artists



- **2,316** Training
- **334** Validation
- **327** Test

2,977 retained

We filtered and deduplicated the source, excluding 36 segments.

Split by artist + song identity

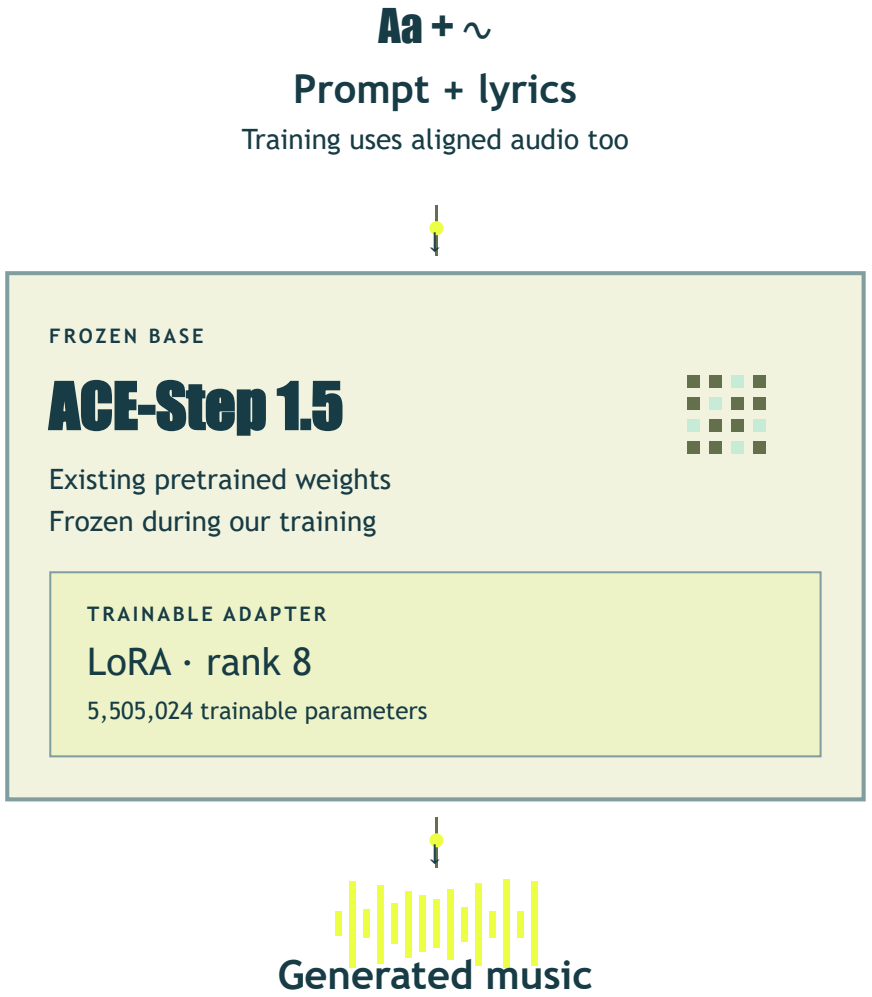
Artists can still occur across splits. This is not an unseen-artist test.

Aligned lyrics + isolated vocals

Aligned text removed a major preparation bottleneck. These are isolated vocals, not full arrangements.

We accessed Kazakh Songs ASR through our approved Hugging Face account, pinned the dataset revision, and prepared it locally. Gated academic-use recordings are not bundled here.

WHAT WE STARTED WITH.
WHAT WE TRAINED.



04 / A COMPLETED LOCAL RUN

MANY RUNS.
ONE WORKING PIPELINE.

300+ min
estimated across training experiments

66.3 min
logged final training run

3,088
updates in that final run



2,316 new clips, each used once
+ 772 updates over nine original music crops

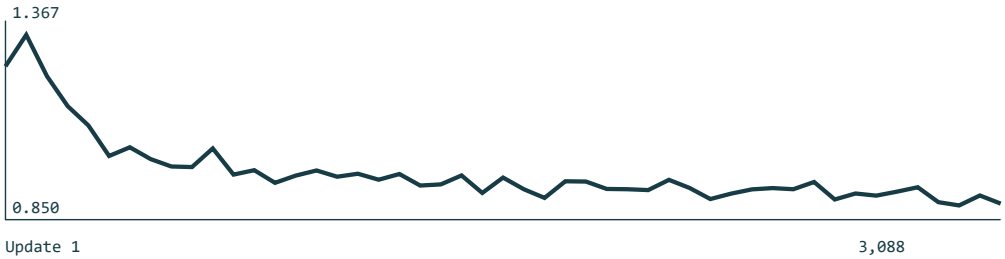
0 held-out clip IDs in the training log
Local machines · frozen base · trained adapter

300+ minutes is our retrospective estimate, not a sum verified from complete logs. The final run is measured separately; its timer excludes preparation, loading and generation.

FROM THE ACTUAL RUN / OPEN THE EVIDENCE

RECORDED STEPS. SAVED MODEL WEIGHTS.

TRAINING LOSS / 64-UPDATE WINDOW MEANS



Each point summarizes recorded optimization steps. A lower value does not establish better music.

VERIFIED LOCAL ARTIFACTS

3,088 logged updates

2,316 distinct new training clips

4 saved checkpoints with different hashes

`Final adapter SHA-256`
`8e0dde48b694…91e561c7`

Open actual training log ↗ Verified checkpoint hashes ↗ Training source ↗ Adapter configuration ↗

We built the preparation and comparison workflow around ACE-Step. The log and saved weights demonstrate adapter training; they do not show foundation-model training from scratch.

THE ENGINEERING / MORE THAN PRESSING GENERATE

THE HARD PARTS.
DATA, LANGUAGE, MEMORY.

01

Checking every source

We found online videos and public datasets, checked provenance and usage terms, extracted suitable audio crops, then checked decoding, silence, clipping and duplicates.

02

Making vocals usable

Automatic lyric drafts were not enough. Singing needs the right words in the right crop. Kazakh text had to survive processing unchanged.

03

Fitting the laptop

We used four-bit base weights, a small adapter and gradient checkpointing to fit the final run on a 4 GB GPU.

We estimate 300+ minutes across training experiments. Listening, source research, filtering, preparation, troubleshooting and generation added further work; total project time was not measured.

THE OUTPUT / GENERATED LOCALLY

FROM OUR PROMPT. TO GENERATED MUSIC.

Same prompt, seed and 30-second duration.
Average playback levels matched.
Prerecorded AI-generated audio.

REFERENCE

Pretrained

Our baseline generation



30-second prerecorded demonstration · play in the website

ADAPTED OUTPUT

25% strength

Our adapted output



30-second prerecorded demonstration · play in the website

These outputs demonstrate the working generation pipeline. The adapter is applied at 25% strength for this example; this is a generation setting, not a quality ranking.

MODEL CHOICE / A PRACTICAL INPUT FORMAT

DESCRIBE THE MUSIC. ADD KAZAKH WORDS.

ALTERNATIVE WE INVESTIGATED

VocalRender: score-driven singing

Requires words, MIDI pitches and note durations, plus tempo. Its documented datasets include Mandarin singing. It produces the vocal line; accompaniment needs a separate path.

For our recordings: this would require detailed note and syllable annotation.

OUR IMPLEMENTED PATH

ACE-Step: prompt + lyrics → music

We supply a description, Kazakh lyrics and duration. The model generates singing and accompaniment. We did not write a note-by-note score.

Our configuration: 30 seconds, fixed seed, 8 generation steps. BPM and key left unspecified; automatic prompt rewriting disabled.

ACTUAL DEMO PROMPT

“Kazakh folk song, solo male vocal, simple acoustic accompaniment, slow tempo, intimate, clean recording”

Open actual prompt and settings ↗ Model-choice record ↗

OUR LOCAL BUILD.
A KAZAKH MUSIC PROTOTYPE.

What we demonstrated

We assembled and checked the data, built the local workflow, trained and saved an adapter, and generated Kazakh-lyrics music examples.

What comes next

More licensed vocal and accompaniment data, native-speaker lyric and pronunciation checks, and broader output evaluation.

YELDAR · AISHA · DIMASH

Questions? 2–3 minutes

Kazakh Songs ASR ↗ ACE-Step 1.5 ↗ Our code & evidence ↗ Local evidence

Our team: Yeldar, Aisha and Dimash. Preparation, training and generation ran on local computers. Hugging Face supplied downloads; ACE-Step supplied the pretrained foundation. This is an adaptation prototype, not a foundation model trained from scratch.